AIM:Energy-Aware MAC Protocol for WSN
Implement duty-cycling MAC protocol and evaluate energy conservation in sensor networks.
CODE:energy_mac.tcl

```
# ============================================================
# Energy-Aware MAC Protocol for WSN
# ============================================================

set ns [new Simulator]

# Trace Files
set tr [open energy.tr w]
$ns trace-all $tr

set nf [open energy.nam w]
$ns namtrace-all-wireless $nf 500 500

# Topography
set topo [new Topography]
$topo load_flatgrid 500 500

create-god 5

# Channel
set chan [new Channel/WirelessChannel]

# Node Configuration
$ns node-config \
    -adhocRouting DSDV \
    -llType LL \
    -macType Mac/802_11 \
    -ifqType Queue/DropTail/PriQueue \
    -ifqLen 50 \
    -antType Antenna/OmniAntenna \
    -propType Propagation/TwoRayGround \
    -phyType Phy/WirelessPhy \
    -channel $chan \
    -topoInstance $topo \
    -agentTrace ON \
    -routerTrace ON \
    -macTrace ON \
    -movementTrace ON \
    -energyModel EnergyModel \
    -initialEnergy 10 \
    -txPower 2.0 \
    -rxPower 1.0 \
    -idlePower 0.5

# Create Nodes
```

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```
for {set i 0} {$i < 5} {incr i} {
    set node_(i)[ns node]
}

# Node Positions
$node_(0) set X_ 50
$node_(0) set Y_ 250

$node_(1) set X_ 150
$node_(1) set Y_ 250

$node_(2) set X_ 250
$node_(2) set Y_ 250

$node_(3) set X_ 350
$node_(3) set Y_ 250

$node_(4) set X_ 450
$node_(4) set Y_ 250

for {set i 0} {$i < 5} {incr i} {
    node_(i) set Z_ 0
    $ns initial_node_pos node_(i) 20
}

# UDP Source
set udp0 [new Agent/UDP]
$ns attach-agent $node_(0) $udp0

# UDP Sink
set null0 [new Agent/Null]
$ns attach-agent $node_(4) $null0

$ns connect $udp0 $null0

# CBR Traffic
set cbr0 [new Application/Traffic/CBR]
$cbr0 set packetSize_ 1000
$cbr0 set interval_ 0.02
$cbr0 attach-agent $udp0

# Start Traffic
nsat1.0"cbr0 start"
nsat20.0"cbr0 stop"

# Finish Procedure
proc finish {} {
    global ns tr nf node_
```

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```
    puts "\n===== ENERGY REPORT ====="

    for {set i 0} {$i < 5} {incr i} {
        puts "Node iRemainingEnergy = [node_($i) energy] J"
    }

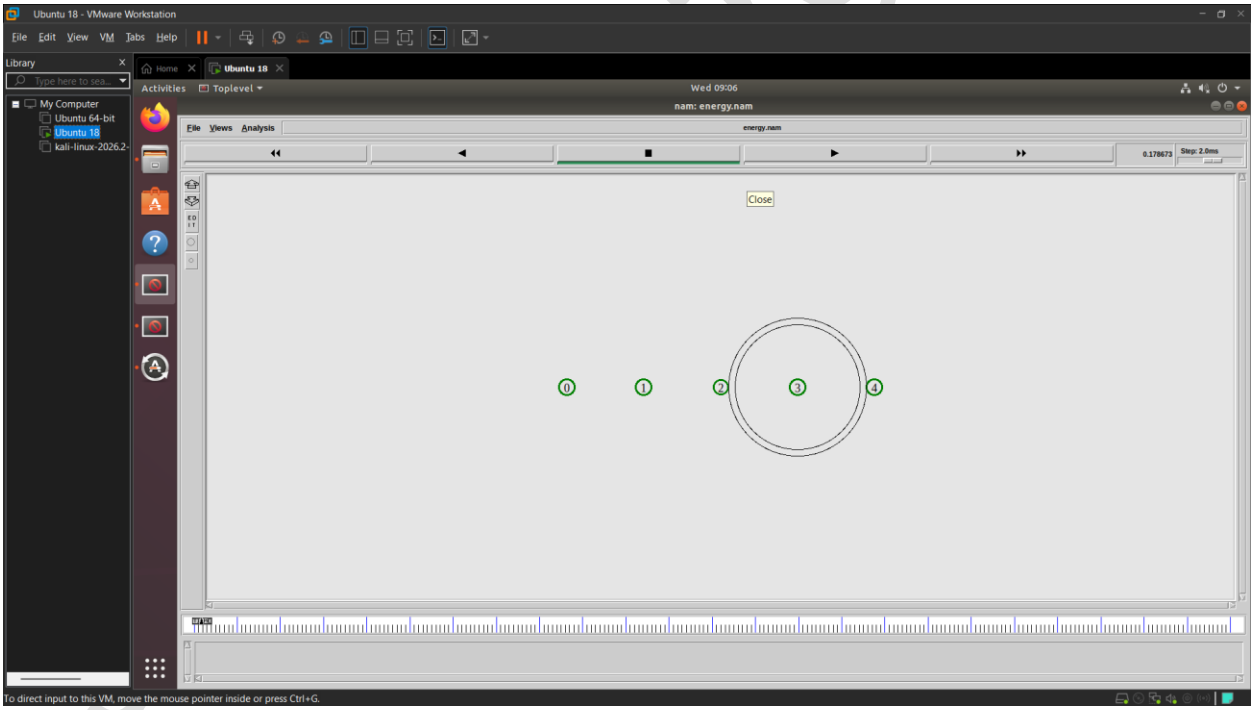
    close $tr
    close $nf

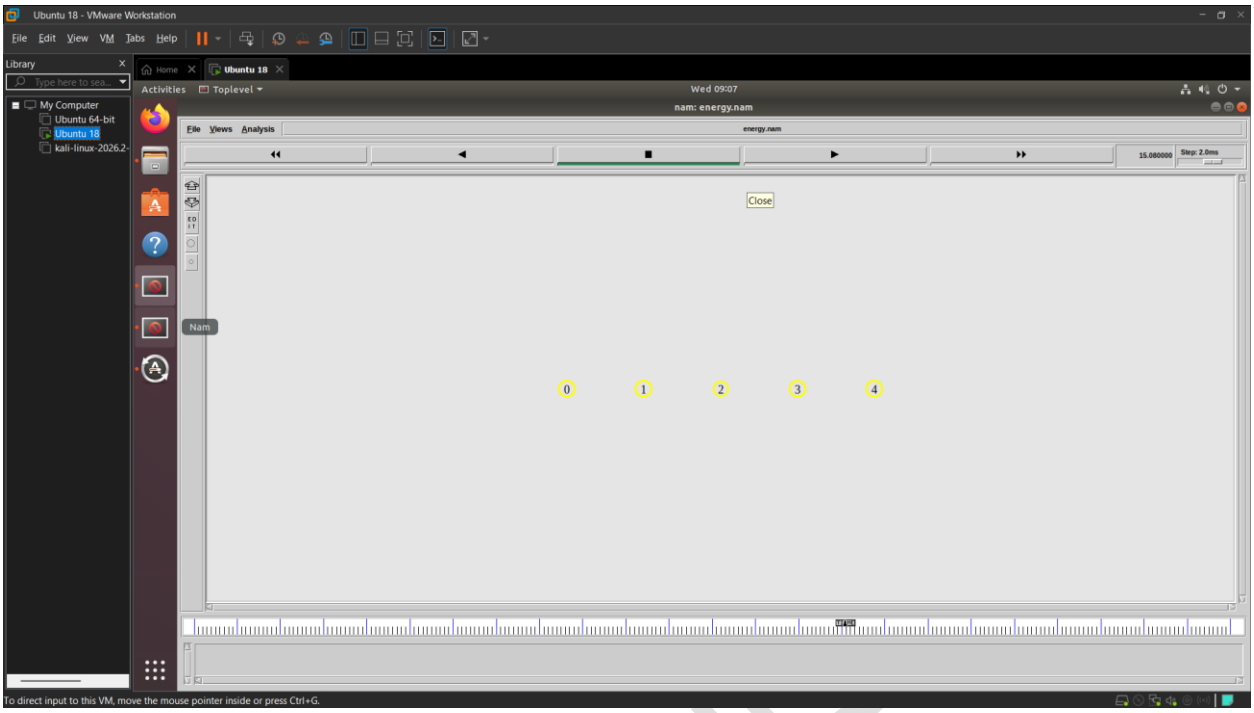
    exec nam energy.nam &
    exit 0
}

$ns at 25.0 "finish"

puts "Simulation Started..."
$ns run
```

OUTPUT:





Observation

- Five sensor nodes were deployed in a linear topology.
- Continuous CBR traffic was generated from Node 0 to Node 4.
- Energy decreased during packet transmission and reception.
- Nodes involved in forwarding packets consumed more energy than idle nodes.
- Remaining energy values were displayed after simulation completion.

Result

The energy-aware WSN simulation was successfully executed. Energy consumption was observed for all sensor nodes, and the remaining energy levels were measured after data transmission.

Jeetesh Shi